Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

Section A **3*10=30**

1)	Analyze the differences between an Eulerian graph and a Hamiltonian graph. Provide an example of a graph that satisfies both conditions.	**3**	(CO1, L4)
2)	Discuss the use of graph theory in scheduling final exams at an educational institution.	**3**	(CO2, L2)
3) a)	Examine a simple graph with 5 vertices and determine if it can have exactly 6 edges. Justify your answer.	**3**	(CO3, L4)
	OR		
b)	Examine the conditions for a graph to be Hamiltonian but not Eulerian. Provide an example of such a graph.	**3**	(CO3, L4)
4)	Analyze and differentiate between a tree and a spanning tree in graph theory.	**3**	(CO1, L4)
5)	For a simple graph with 8 vertices, determine the number of edges required to ensure that every vertex has an even degree. Explain your reasoning.	**3**	(CO2, L2)
7)	Discuss how the number of edges in a simple graph and its complement is related, and determine if this relationship holds for a graph with 5 vertices.	**3**	(CO1, L2)
8) a)	Define a spanning tree and demonstrate how to construct one for the graph given below by removing exactly two edges.	**3**	(CO2, L3)
	OR		
b)	Construct a Huffman coding scheme for the following symbols with their respective frequencies and determine the average number of bits per symbol encoded. A: 0.15, B: 0.20, C: 0.30, D: 0.10, E: 0.25. Show your steps in constructing the coding tree.	**3**	(CO2, L3)
9)	Construct a simple graph with 5 vertices such that each vertex has an odd degree. Determine the number of edges in your constructed graph.	**3**	(CO3, L3)

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10)	Discuss how many edges a simple graph with 6 vertices can have at most, and explain your reasoning.	**3**	(CO1, L2)

Section B **7*3=21**

11)	Analyze the role of graph theory in scheduling final exams at a university. Specifically, (i) describe how vertex coloring can be used to minimize conflicts among exam timeslots and courses. (ii) Discuss the limitations of this approach considering practical constraints such as classroom availability and student preferences.	**7**	(CO2, L4)
12) a)	A simple graph G has 8 vertices and 12 edges. Determine (i) whether it is possible for G to have exactly 4 vertices of odd degree, and justify your answer using a relevant theorem from Graph Theory. (ii) Construct an example of such a graph G if the condition in part (i) can be satisfied, or explain why it cannot be constructed.	**7**	(CO3, L3)
	OR		
b)	Construct a simple graph with 6 vertices and determine the minimum number of edges required to ensure that each vertex has an even degree. Demonstrate your answer by drawing the graph and calculating the degrees of each vertex.	**7**	(CO3, L3)
13)	Discuss the concept of a complete graph and its properties. (i) Determine whether a simple graph with 6 vertices can be both complete and have exactly 9 edges. Justify your answer. (ii) Given two graphs G1 and G2, where G1 is a complete graph on 4 vertices and G2 is a complete bipartite graph K3,3, explain the differences between their structures and number of edges.	**7**	(CO1, L2)

Section C **7*4=28**

14)	Design a graph-based scheduling system to minimize conflicts during final exams at a university. Your solution should consider the following aspects: (i) Construct a graph representing courses and their dependencies, where vertices are courses and edges represent prerequisites. (ii) Determine an optimal schedule for 5 final exams, ensuring no student has two exams at the same time. (iii) Discuss how vertex coloring can be used to identify potential scheduling conflicts in your graph.	**7**	(CO2, L3)
15)	Determine the number of edges in a simple graph with 8 vertices where each vertex has an even degree, and construct such a graph if possible. Demonstrate your solution by explaining the process used to ensure all vertices have even degrees.	**7**	(CO3, L3)
16) a)	Consider a simple graph G with 8 vertices and 15 edges. Determine (i) whether it is possible for this graph to have exactly 4 vertices of odd degree, (ii) if such a graph exists, provide an example by constructing its adjacency matrix, or prove that no such graph can exist. Justify your answer using relevant theorems from Graph Theory.	**7**	(CO1, L4)
	OR		
b)	Analyze the properties of simple graphs with an even number of vertices and determine whether such graphs can have a unique perfect matching. (i) State the necessary conditions for a graph to have a perfect matching. (ii) Construct a simple graph with 6 vertices that has a unique perfect matching, if possible, or prove why it cannot exist.	**7**	(CO1, L4)

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17)	Analyze a simple graph G with 8 vertices and determine whether it can have exactly 15 edges. If possible, provide an example of such a graph. If not, justify why such a configuration is impossible using concepts from graph theory.	**7**	(CO2, L4)

Section D **7*3=21**

18)	Analyze the differences between Hamiltonian and Eulerian graphs using examples. Additionally, investigate how the concepts of these graphs can be applied to solve practical problems in network design.	**7**	(CO3, L4)
19) a)	Analyze whether a simple graph with 6 vertices can exist where each vertex has an odd degree and explain your reasoning using relevant graph theory concepts. (i) Differentiate between a complete graph and a regular graph, providing examples to support your answer. (ii) Consider two simple graphs G1 and G2 with adjacency matrices as follows: G1 = [0 1 0; 1 0 1; 0 1 0] G2 = [0 1 1; 1 0 1; 1 1 0] Examine if G1 and G2 are isomorphic. Justify your conclusion with a detailed comparison of their structural properties.	**7**	(CO1, L4)
	OR		
b)	Examine a simple graph with 6 vertices and analyze whether it can have exactly 5 edges without being a tree. Justify your answer using relevant properties of graphs. (i) Determine if such a graph could exist, providing reasoning based on necessary conditions for trees. (ii) Construct an example of such a graph or prove that it is impossible, by applying the Handshaking Lemma and Euler's formula for planar graphs.	**7**	(CO1, L4)
20)	Construct a graph that represents the scheduling of final exams for three subjects: Mathematics, Physics, and Chemistry in such a way that no two exams are scheduled at the same time slot if there exists an overlap in student enrollment (i.e., if a student is enrolled in both subjects). Demonstrate your solution by finding a valid schedule or color coding scheme for the exam slots using a minimum number of colors. Additionally, show why your coloring scheme meets the criteria.	**7**	(CO2, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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